

# Introduction to Limits



## Evaluating limits

- 1 Evaluate  $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$  by first simplifying the expression.
  - 2 Evaluate by direct substitution where possible:
    - a  $\lim_{x \rightarrow 3} (x^2 + 2x - 1)$
    - b  $\lim_{x \rightarrow 1} \frac{x+5}{x+1}$
  - 3 Using a table of values near  $x = 0$  (in radians), estimate  $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ .
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## One-sided limits and limits at infinity

- 4 Let  $f(x) = \frac{|x|}{x}$ . Find:
  - a  $\lim_{x \rightarrow 0^+} f(x)$
  - b  $\lim_{x \rightarrow 0^-} f(x)$
  - c  $\lim_{x \rightarrow 0} f(x)$
- 5 Evaluate each limit at infinity:
  - a  $\lim_{x \rightarrow \infty} \frac{3x^2 + 5}{x^2 - 2}$
  - b  $\lim_{x \rightarrow \infty} \frac{2x+1}{x^2 + 4}$
- 6 The graph of  $f(x) = \frac{1}{x-2} + 3$  has asymptotes  $x = 2$  and  $y = 3$ . Express each asymptote as a limit statement.